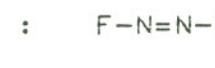
PHYSICAL CHEMISTRY CUMULATIVE EXAM

December 3, 1994

There are three plausible geometries for difluorodiazine,

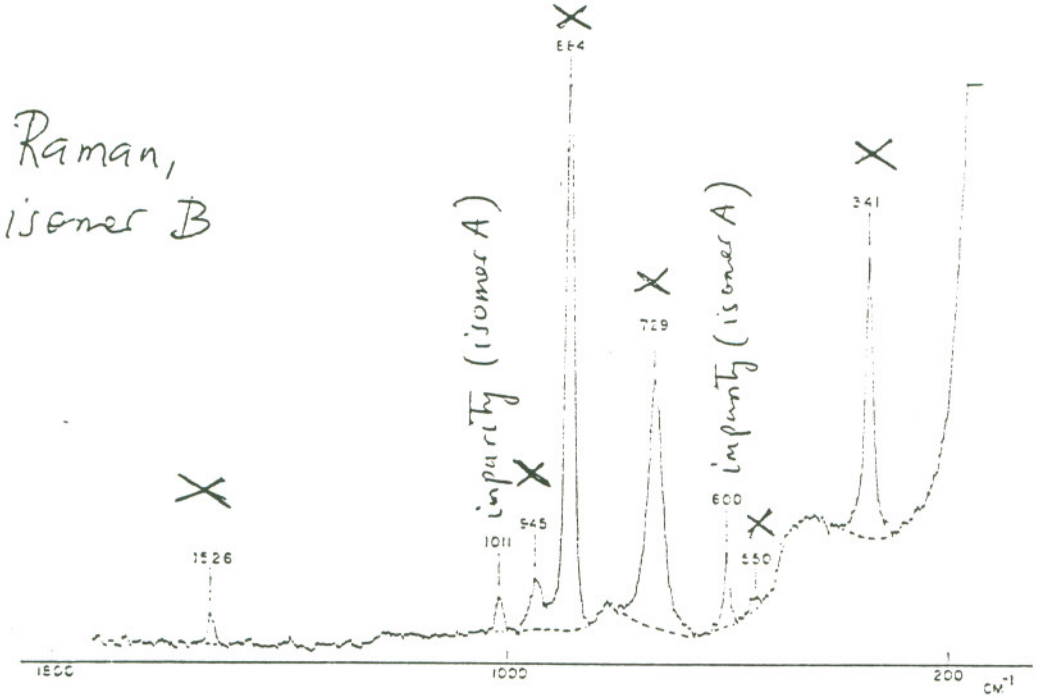
N_2F_2: linear : F–N=N–F ; cis :  , and trans: .

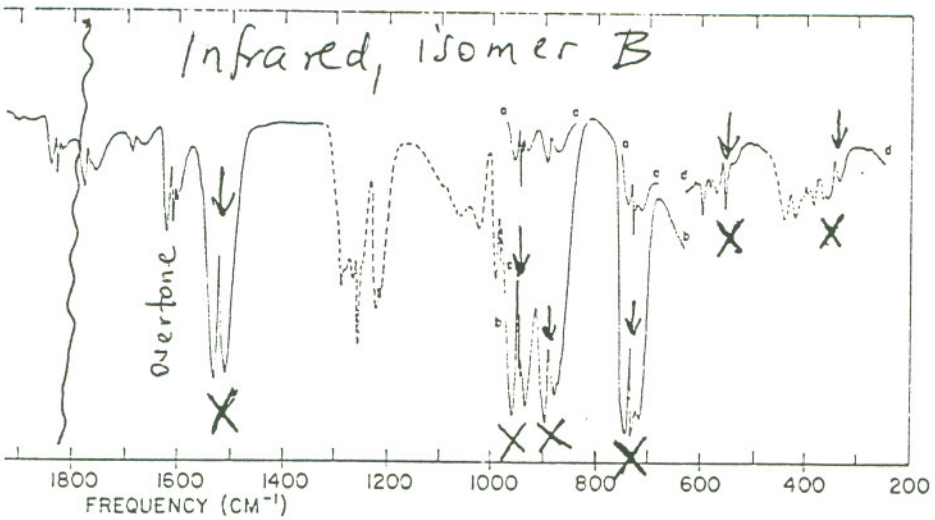
1. What is the symmetry point group of these three forms?

2. How many vibrational degrees of freedom are there in each of these forms?

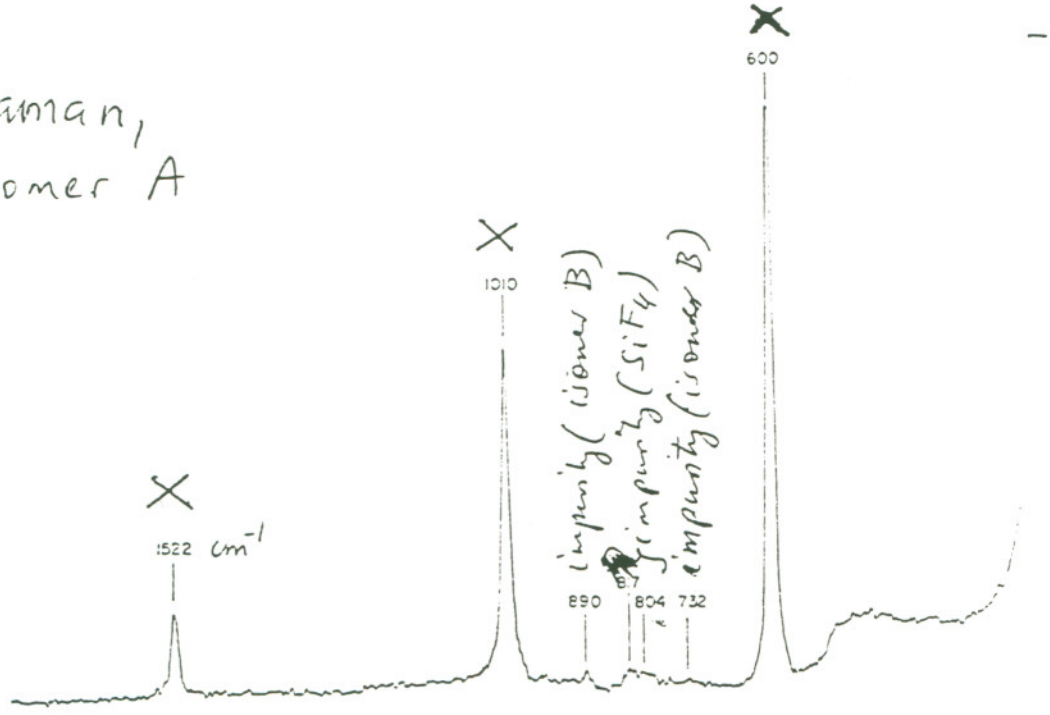
3. Using the enclosed character tables, determine the number of Raman-active and infra-red active fundamental vibrations for these three possible isomers of N_2F_2.

4. Crude N_2F_2 is a mixture of two isomers which can be separated by destillation and gas chromatography. The observed Raman and infra-red spectra of these two isomers, A and B, are enclosed [see King and Overend, Spectrochim. Acta A, 23, 61 (1967); 22, 689 (1966)]. Vibrational bands which are probably fundamentals are denoted by an X. Impurity bands are also designated as such. Using the preceding theory, assign isomer A to one of the three possible geometries (linear, cis, trans). Do the same for isomer B. **Include a detailed justification for your choice.**


Raman,
isomer B
664
729
341
1526
1011
545
600
550
impurity (isomer A)
impurity (isomer A)
1500
1000
200
cm⁻¹


Infrared, isomer B
overtone
1800
1600
1400
1200
1000
800
600
400
200
FREQUENCY (CM⁻¹)

Raman, isomer A



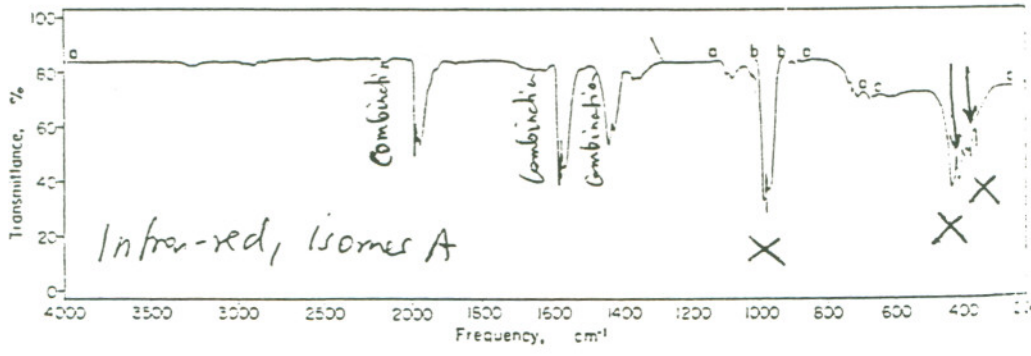


TABLE X-3. CHARACTER TABLES OF $\mathfrak{C}_s$, $\mathfrak{C}_i$, AND THE CYCLIC GROUPS $\mathfrak{C}_n$
($n = 2, 3, 4, 5, 6$)

$\mathfrak{C}_s$	E	σ_h		
A'	1	1	$T_x, T_y; R_z$	$\alpha_{xx}, \alpha_{yy}, \alpha_{zz}, \alpha_{xy}$
A''	1	-1	$T_z; R_x, R_y$	α_{yz}, α_{xz}

$\mathfrak{C}_i$	E	i		
A_g	1	1	R	α
A_u	1	-1	T	

$\mathfrak{C}_2$	E	C_2		
A	1	1	$T_z; R_z$	$\alpha_{xx}, \alpha_{yy}, \alpha_{zz}, \alpha_{xy}$
B	1	-1	$T_x, T_y; R_x, R_y$	α_{yz}, α_{xz}

TABLE X-5. CHARACTER TABLES OF THE GROUPS
$\mathfrak{C}_{nv}$ ($n = 2, 3, 4, 5, 6$)

$\mathfrak{C}_{2v}$	E	C_2	$\sigma_v(xz)$	$\sigma_v(yz)$		
A_1	1	1	1	1	T_z	$\alpha_{xx}, \alpha_{yy}, \alpha_{zz}$
A_2	1	1	-1	-1	R_z	α_{xy}
B_1	1	-1	1	-1	$T_x; R_y$	α_{xz}
B_2	1	-1	-1	1	$T_y; R_x$	α_{yz}

$\mathfrak{C}_{3v}$	E	$2C_3$	$3\sigma_v$		
A_1	1	1	1	T_z	$\alpha_{xx} + \alpha_{yy}, \alpha_{zz}$
A_2	1	1	-1	R_z	
E	2	-1	0	$(T_x, T_y); (R_x, R_y)$	$(\alpha_{xx} - \alpha_{yy}, \alpha_{xy}); (\alpha_{yz}, \alpha_{xz})$

TABLE X-6. CHARACTER TABLES OF THE GROUPS $\mathfrak{C}_{nh}$
($n = 2, 3, 4, 5, 6$)

$\mathfrak{C}_{2h}$	E	C_2	i	σ_h		
A_g	1	1	1	1	R_z	$\alpha_{xx}, \alpha_{yy}, \alpha_{zz}, \alpha_{xy}$
B_g	1	-1	1	-1	R_x, R_y	α_{yz}, α_{xz}
A_u	1	1	-1	-1	T_z	
B_u	1	-1	-1	1	T_x, T_y	

TABLE X-11. CHARACTER TABLES OF THE GROUPS, $\mathfrak{C}_{\infty v}$ and $\mathfrak{D}_{\infty h}$, FOR
LINEAR MOLECULES

$\mathfrak{C}_{\infty v}$	E	$2C_\infty^\phi$	...	$\infty\sigma_v$		
$A_1 \equiv \Sigma^+$	1	1	...	1	T_z	$\alpha_{xx} + \alpha_{yy}, \alpha_{zz}$
$A_2 \equiv \Sigma^-$	1	1	...	-1	R_z	
$E_1 \equiv \Pi$	2	$2\cos\phi$	...	0	$(T_x, T_y); (R_x, R_y)$	$(\alpha_{yz}, \alpha_{xz})$
$E_2 \equiv \Delta$	2	$2\cos 2\phi$	...	0		$(\alpha_{xx} - \alpha_{yy}, \alpha_{xy})$
$E_3 \equiv \Phi$	2	$2\cos 3\phi$	...	0		
...	...	...	...	...		

$\mathfrak{D}_{\infty h}$	E	$2C_\infty^\phi$	...	$\infty\sigma_v$	i	$2S_\infty^\phi$	...	∞C_2		
Σ_g^+	1	1	...	1	1	1	...	1		$\alpha_{xx} + \alpha_{yy}, \alpha_{zz}$
Σ_g^-	1	1	...	-1	1	1	...	-1	R_z	
Π_g	2	$2\cos\phi$	...	0	2	$-2\cos\phi$	...	0	(R_x, R_y)	$(\alpha_{yz}, \alpha_{xz})$
Δ_g	2	$2\cos 2\phi$	...	0	2	$2\cos 2\phi$	...	0		$(\alpha_{xx} - \alpha_{yy}, \alpha_{xy})$
...	...	...	...	...	...	...	...	...		
Σ_u^+	1	1	...	1	-1	-1	...	-1	T_z	
Σ_u^-	1	1	...	-1	-1	-1	...	1		
Π_u	2	$2\cos\phi$	...	0	-2	$2\cos\phi$	...	0	(T_x, T_y)	
Δ_u	2	$2\cos 2\phi$	...	0	-2	$-2\cos 2\phi$	...	0		
...	...	...	...	...	...	...	...	...		